

In three dimensions the mass gap is the positivity of one function at infinity

For pure Yang–Mills in $d = 3$ (two space dimensions) the coupling g^2 is itself a mass, so the finite-volume gap of the continuum theory on a periodic square torus of side L has the exact form

$$\Delta(L) = g^2 \hbar^2 c f(x), \quad x = g^2 \hbar L \text{ (dimensionless),}$$

by dimensional analysis alone, and the Jaffe–Witten conjecture for $d = 3$ is the single statement

$$\lim_{x \rightarrow \infty} f(x) = C \in (0, \infty), \quad m = C g^2 \hbar^2 / c.$$

The small- x end of f is known from the constant-mode sector: with two spatial components the zero-mode Hamiltonian is C133 with $D = 2$, $m_{\text{eff}} = L^2/(g^2 c)$ and $g_B^2 = L^2 c / g^2$, giving $\Delta_0(L) = \delta_1^{(2)} g^2 \hbar^2 c x^{-2/3}$, that is $f(x) = \delta_1^{(2)} x^{-2/3} [1 + o(1)]$ as $x \rightarrow 0$. On the lattice, the strong-coupling theorem of the previous note with $\nu = 2$ gives a gap $\geq \gamma_2(g_{\text{lat}}^2/2) C_2 \hbar c / a$ uniform in the lattice size, and the continuum limit is the statement that $\delta_\infty(g_{\text{lat}})/g_{\text{lat}}^2$ stays bounded below as $g_{\text{lat}}^2 = g^2 \hbar a \rightarrow 0$: the gap-to-coupling ratio is bounded below at both ends of the coupling axis, and the conjecture is its positivity in between. Contrast $d = 4$, where the ratio must pass from a power law $\propto g^2$ at strong coupling to the essential singularity $e^{-1/(2b_0 g^2)}$ at weak coupling. The three-dimensional problem is therefore the natural first target: it has no transmutation, a single scaling variable, and both ends of f in hand. The expected value $C = N/(2\pi)$ (Karabali–Nair, physics argument; metadata, B78) is a prediction to be proved, not an input. Nothing here is promoted.

1. Units and the one-variable form

With $S = \frac{1}{4g^2} \int F_{\mu\nu}^a F^{a\mu\nu} d^3x$ and the quantization $e^{iS/\hbar}$, the dimensional census of G07 §4 gives $[1/g^2] = \text{action-length in } d = 3$, so $g^2 \hbar = 1/\ell_g$ is an inverse length, $\mu := g^2 \hbar^2 / c = \hbar/(c\ell_g)$ is a mass, and $\varepsilon_3 := \mu c^2 = g^2 \hbar^2 c$ is the unique energy formed from $(\hbar, c, g)$. The continuum theory on the periodic torus $T_L^2 \times \mathbb{R}$ has the fixed constants $\hbar, c, g$ and the box size L , and a gap $\Delta(L)$, when the finite-volume theory exists, depends on them only. By the floor/unit theorem C131 applied to the observable “energy”, $\Delta(L)$ is ε_3 times a function of the single dimensionless combination $x = L/\ell_g = g^2 \hbar L$, because $(\hbar, c, g)$ span exactly one energy and L adds exactly one dimensionless ratio. Setting $f(x) = \Delta(L)/\varepsilon_3$:

$$\Delta(L) = \varepsilon_3 f(x), \quad \varepsilon_3 = g^2 \hbar^2 c, \quad x = g^2 \hbar L.$$

The infinite-volume gap, if it exists, is $mc^2 = \varepsilon_3 \lim_{x \rightarrow \infty} f(x)$, and Jaffe–Witten’s $0 < m < \infty$ is $0 < \lim f < \infty$. Vacuum-energy renormalization, the only ultraviolet divergence of the superrenormalizable theory relevant here, shifts E_0 and E_1 equally and leaves f untouched.

2. The small- x end from the constant modes

Temporal gauge on T_L^2 leaves two constant spatial components a_1, a_2 in the Lie algebra. With $S = \frac{1}{4g^2} \int F_{\mu\nu}^a F^{a\mu\nu} d^2x c dt$, $F_{0i} = c^{-1} \dot{a}_i$ and $F_{12} = \vec{a}_1 \times \vec{a}_2$, the constant-mode Lagrangian is $\frac{L^2}{2g^2 c} |\dot{\vec{a}}|^2 - \frac{L^2 c}{2g^2} |\vec{a}_1 \times \vec{a}_2|^2$ and

$$H_0 = \frac{g^2 c}{2L^2} \sum_{i=1}^2 |\vec{p}_i|^2 + \frac{L^2 c}{2g^2} |\vec{a}_1 \times \vec{a}_2|^2, \quad [a_i^a, p_j^b] = i\hbar \delta_{ij} \delta^{ab},$$

which is C133 with $D = 2$, $m_{\text{eff}} = L^2/(g^2 c)$ and $g_B^2 = L^2 c / g^2$; $\hbar$ enters only through g^2 and the commutator. Its energy unit is

$$\varepsilon = \hbar^{4/3} g_B^{2/3} m_{\text{eff}}^{-2/3} = \hbar^{4/3} \left(\frac{L^2 c}{g^2} \right)^{1/3} \left(\frac{L^2}{g^2 c} \right)^{-2/3} = \hbar^{4/3} c g^{2/3} L^{-2/3} = \frac{\hbar c}{L} \left(\frac{L}{\ell_g} \right)^{1/3} = \varepsilon_3 x^{-2/3},$$

using $g^{2/3}\hbar^{1/3} = \ell_g^{-1/3}$ and $\hbar c/L = \varepsilon_3/x$. So the constant-mode gap is $\Delta_0(L) = \delta_1^{(2)}\varepsilon_3 x^{-2/3}$ with $\delta_1^{(2)} > 0$ the pure number of C133 for $D = 2$, and

$$f(x) = \delta_1^{(2)} x^{-2/3} [1 + o(1)], \quad x \rightarrow 0,$$

where $o(1)$ stands for the coupling to the nonzero momentum modes, controlled perturbatively by the small-volume expansion in the same way as in $d = 4$ (Lüscher's method; the $d = 3$ expansion parameter is $x^{1/3}$ rather than $g^{2/3}$, since x is the only parameter). The claim $f \rightarrow \infty$ as $x \rightarrow 0$ is the statement that the small box is gapped with a gap growing like $L^{-2/3}$, slower than the $1/L$ of a free particle in a box: the zero-point mechanism of G08 gives a softer confinement than a hard wall.

3. The lattice route in $d = 3$ and the strong-coupling end

On the periodic square lattice $(a\mathbb{Z}/N_s a\mathbb{Z})^2$ the Kogut–Susskind Hamiltonian is

$$H = \frac{\hbar c}{a} \left[\frac{g_{\text{lat}}^2}{2} \sum_{\ell} (-\Delta_{\ell}) + \frac{2}{g_{\text{lat}}^2} \sum_p (N - \text{Re tr } U_p) \right], \quad g_{\text{lat}}^2 = g^2 \hbar a = \frac{a}{\ell_g},$$

with g_{lat} dimensionless. Theorem 1 of the obligations map holds verbatim in two dimensions: $\Delta_{a,L} = (\hbar c/a) \delta(g_{\text{lat}}; N_s, G) > 0$ with a unique physical ground state. The strong-coupling theorem holds with $\nu = 2$, $\Lambda_0 = \{0, 1\}^2$, two links and one plaquette per site:

$$\beta = \frac{4}{g_{\text{lat}}^4 C_2} \cdot 2N = \frac{16N^2}{g_{\text{lat}}^4 (N^2 - 1)}, \quad \Delta_{a,L} \geq \gamma_2 \frac{g_{\text{lat}}^2}{2} C_2 \frac{\hbar c}{a},$$

for $g_{\text{lat}} \geq g_0^{(2)}(N) = (16N^2 / [(N^2 - 1)\beta_*(2, \{0, 1\}^2)])^{1/4}$, uniformly in N_s , by the same hypothesis check.

The continuum limit is $a \rightarrow 0$ at fixed g^2 , L , that is $g_{\text{lat}}^2 \rightarrow 0$ with $N_s = L/a$ and $x = g^2 \hbar L = g_{\text{lat}}^2 N_s$ fixed. The lattice gap in units of ε_3 is

$$\frac{\Delta_{a,L}}{\varepsilon_3} = \frac{\hbar c}{a g^2 \hbar^2 c} \delta = \frac{\delta(g_{\text{lat}}; N_s, G)}{g_{\text{lat}}^2},$$

so the continuum finite-volume gap exists iff $\delta(g_{\text{lat}}; x/g_{\text{lat}}^2, G)/g_{\text{lat}}^2 \rightarrow f(x)$ as $g_{\text{lat}} \rightarrow 0$, and the infinite-volume mass is

$$\frac{mc^2}{\varepsilon_3} = \lim_{g_{\text{lat}} \rightarrow 0} \frac{\delta_{\infty}(g_{\text{lat}})}{g_{\text{lat}}^2}, \quad \delta_{\infty}(g_{\text{lat}}) = \liminf_{N_s \rightarrow \infty} \delta(g_{\text{lat}}; N_s, G).$$

At strong coupling the same ratio is bounded below by $\gamma_2 C_2/2$. The $d = 3$ conjecture on the lattice route is therefore:

T3₍₃₎. The function $c(g_{\text{lat}}) = \delta_{\infty}(g_{\text{lat}})/g_{\text{lat}}^2$, which satisfies $c \geq \gamma_2 C_2/2$ for $g_{\text{lat}} \geq g_0^{(2)}$, stays bounded below by a positive constant for all $g_{\text{lat}} > 0$ and has a finite limit C as $g_{\text{lat}} \rightarrow 0$.

The abelian contrast in $d = 3$. For compact $U(1)$ in three dimensions Göpfert and Mack (abstract, B78) prove a gap at every coupling, so T2' holds for the abelian group here, and their Debye mass $m_D^2 = (2\beta/a^3)e^{-\beta v(0)/2}$, $\beta = 4\pi^2/e_{\text{lat}}^2$, makes $c(g_{\text{lat}}) = \delta_{\infty}/g_{\text{lat}}^2$ vanish with an essential singularity as $g_{\text{lat}} \rightarrow 0$: the continuum photon is massless. In $d = 3$ the abelian/non-abelian distinction is therefore the limit of the ratio, zero against a positive C , with positivity at each coupling shared by both. Everything in this note up to here is a collection of known facts in one normalization; the only statement of this programme's own is the reduction to f and its two ends.

No exponential enters. In $d = 4$ the corresponding ratio $\delta_\infty(g)/g^2$ is bounded below at strong coupling by the same theorem and must vanish like $g^{-2}e^{-1/(2b_0g^2)}$ at weak coupling; the two ends of the $d = 4$ problem have different functional forms and the lattice gap must interpolate between them, which is what makes T3 in $d = 4$ a transmutation statement. In $d = 3$ both ends are linear in g_{lat}^2 .

4. What is in hand and what is not

	$d = 3$	$d = 4$
scale of the gap	$\varepsilon_3 = g^2 \hbar^2 c$, classical constant	$\hbar c \Lambda$, generated by transmutation
finite-volume gap	$\varepsilon_3 f(x)$, $x = g^2 \hbar L$	$(\hbar c/L) z(L)$ with $g(L)$ running
small-volume end	$f(x) = \delta_1^{(2)} x^{-2/3} [1 + o(1)]$	$z = \delta_1^{(3)} g(L)^{2/3} [1 + O(g^{2/3})]$
strong-coupling lattice end	$\delta_\infty/g_{\text{lat}}^2 \geq \gamma_2 C_2/2$	$\delta_\infty/g^2 \geq \gamma_3 C_2/2$
conjecture	$\inf_{g_{\text{lat}}} c(g_{\text{lat}}) > 0$, $c \rightarrow C$	$\delta_\infty(g)/(a\Lambda_{\text{lat}}(g)) \rightarrow m/(\hbar c \Lambda_{\text{lat}})$
expected constant	$C = N/(2\pi)$ (Karabali–Nair, physics)	none in closed form
existence	finite volume by stochastic quantisation (Chevyrev review, passage-level companion in docs)	finite-volume UV stability only

Open in $d = 3$: (i) the infinite-volume limit of the finite-volume construction; (ii) any lower bound on f away from the two ends; (iii) the identification of f 's small- x correction with the stochastic construction's estimates. Item (ii) is the conjecture itself, and the one-variable form means that a single inequality $f(x) \geq f_- > 0$ for $x \geq x_0$, proved for the continuum finite-volume theory, together with the existence of $\lim f$, would settle the $d = 3$ mass gap.

5. Consequence for STATE

The $d = 3$ target is now a statement about one function of one variable with both ends known: $f(x) \sim \delta_1^{(2)} x^{-2/3}$ at $x \rightarrow 0$ and $f \rightarrow C$ conjectured at $x \rightarrow \infty$, with the lattice ratio $\delta_\infty/g_{\text{lat}}^2$ bounded below at strong coupling. It replaces the $d = 3$ line of STATE. The next theorem-sized step in $d = 3$ is a lower bound $f(x) \geq f_- > 0$ on an interval $x \in [x_0, x_1]$ beyond the small-volume expansion, for which the candidate tool is a variational or operator-inequality argument on the torus Hamiltonian that keeps the zero-point mechanism of the constant modes while controlling the nonzero modes by their Gaussian part, which is what the weak-coupling lattice bound of STATE step 3 is meant to prepare.